Modelling turbulent and magnetised plasmas at Exascale with AI-assisted simulation tools

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1 Background and significance

1.1 Significance of research project in relation to current knowledge, research-based starting points

Precise and accurate modeling of magnetized plasmas stands as a formidable challenge that has long vexed a broad spectrum of the scientific community, ranging from reactor engineers and space scientists to astrophysicists [3, 24, 36, 6, 2]. This challenge stems from the inability to run direct numerical simulations (DNS), which can faithfully represent both small-scale and large-scale physics due to the spatial and temporal scales that cover many orders of magnitude as well as the extreme conditions prevalent in these environments. Even with the current supercomputing resources, we are able to run only DNS-type models, where the values of diffusivities are elevated by orders of magnitude from their real values. An example of a high-resolution DNS-type simulation produced by our group is shown in Figure 1.

Mean-field models have emerged as a more accessible alternative by simplifying the representation of smaller-scale phenomena through turbulent transport coefficients that couple small-scale physics with large-scale physics [20]. However, their efficacy is compromised by a fundamental gap in our understanding of turbulent transport mechanisms, leading to the reliance on overly simplistic or selectively chosen turbulent effects [7, 46, 40].

To address these challenges, subgrid-scale (SGS) models[1] have emerged as a popular and effective alternative. Implicit Large Eddy Simulations (ILES) employ a class of numerically motivated SGS models, designed for simulating plasmas in a manner maximizing the flow Reynolds numbers, which controls the intensity of turbulence [for a recent review, see 22]. Despite its success in many applications [41, 30], ILES may not be fully suitable for, e.g., solar-like complicated magnetised plasmas [39], where significant differences in behavior have been observed when compared to DNS-type simulations [15, 17, 14, 43]. These discrepancies highlight the need for more sophisticated SGS models. They have been sought with ansatzes stemming from mean-field angular momentum and electrodynamics theories. One

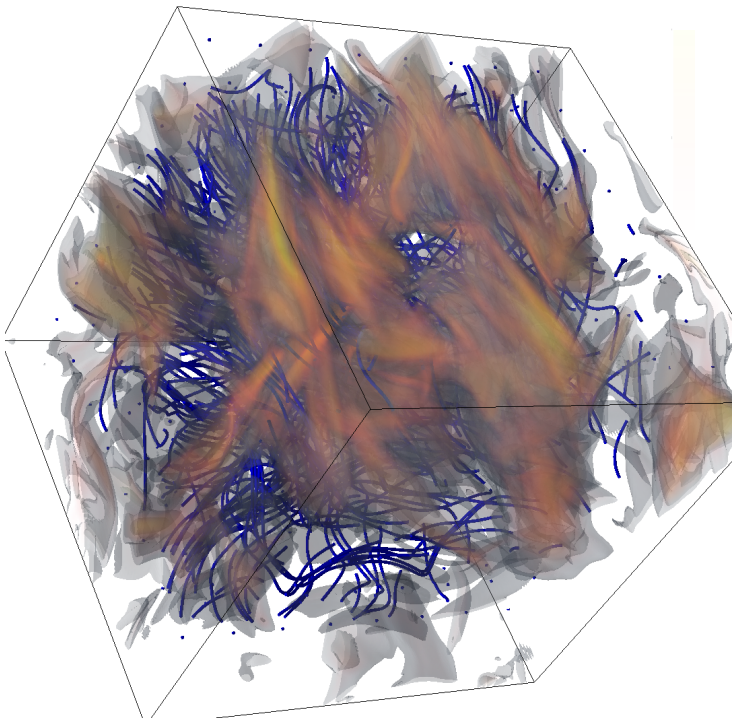

Figure 1: Dynamo-active isothermal forced flow simulated with hybrid MHD simulation tool PC-A. These simulations of the generation and saturation of magnetic field fluctuations, through a so-called small-scale dynamo instability, represent world's largest simulations of their kind [44].

[1]Subgrid-scale models are surrogate models that bridge the gap between large-scale flow behavior and small-scale processes in simulations by approximating the effects of smaller scales, which remain unresolved due to computational constraints. These models aim to provide a sufficiently accurate representation of small-scale physical phenomena, such as turbulent transport, by incorporating their cumulative impact on the larger scales.

approach fits the coefficients to the simulated data using techniques such as singular value decomposition and enhancements [5, 29, 38, 1, 37], another using test field methods [34, 35, 4, 11, 45, 46, 19]. The former suffers from the non-uniqueness of the fitting procedure and hence produces unreliable results [16], and the latter becomes unpracticable for large-scale data due to the huge computational resources required.

The most sophisticated framework proposed so far is the Explicit LES (ELES) approach, which employs physically motivated SGS models to capture better the complexity of flows affected by rotation, density stratification, and magnetic fields. Some ELES-SGS frameworks have been proposed [13, 12], but none of these methods have been rigorously shown to capture the turbulent transport processes that lead to large-scale structure formation in realistic settings, such as the generation of global-scale magnetic structures and oscillatory dynamo modes in the Sun.

Artificial intelligence (AI) and machine learning (ML) models have recently been increasingly employed to address the computational intensity and complexity of traditional simulation methods [10, 18, 42]. These techniques offer significant speedups in computation time and the ability to capture complex physical phenomena through data-driven models [9, 32, 33]. By training on data from DNS-type and even empirical data, these models have the potential to infer underlying physical processes at small scales and reveal new insights into the behavior of plasmas. Despite recent advances, how to model and capture small-scale physics and create effective and scalable AI-assisted ELES models for turbulent and magnetized plasmas remains an open problem.

We propose a novel path forward, solving all the problems described above:

- **GPU-DNS concept**: We will take advantage of GPU-accelerated DNS-type simulations (see Figure 1), enabling the production of unprecedented resolution simulations using the fastest available supercomputer resources, such as the Large Unified Modern Infrastructure (LUMI) and its GPU partition in Finland. These methods have been developed and optimized in-house in our research group, and represent cuttings edge performance and scale-up of high-order turbulence simulations in the world.
- **AI-IN-LOOP concept**: We will undertake the data analysis of the simulations while they are running (known as a 'simulation-in-the-loop' approach), streaming the data between the DNS-type simulation and AI tool, without any necessity to dump the huge system states to the disk, completely avoiding the data analysis bottleneck. This way, we will train deep neural networks (DL) to learn accurate and generalisable descriptions (hereafter denoted subgrid-scale, SGS, models) of the small-scale processes on-the-fly.
- **LES-DNS concept**: We will build next-generation simulation tools on the derived SGS models. SGS models aim to bridge the gap between large-scale behaviour and small-scale processes by approximating the effects of the smaller scales. This project aims at applying the new simulation framework to study an important real-world application, namely convection under solar-like conditions. Such a setup acts as a hydromagnetic dynamo, generating and maintaining small- and large-scale magnetic fields. This system has been studied over 70 years without a definite answer to the question how the Sun generates its oscillatory magnetic field and hence its activity cycle driving space weather phenomena.

Our next-generation simulation tools aim to provide a sufficiently accurate representation of small-scale physical phenomena by incorporating their cumulative impact on larger scales. As a result, these models will no longer need to resolve the smallest-scale features, reducing to computationally much cheaper models and therefore require significantly less data storage. These tools will enable faster and more sustainable progress in various areas of science and engineering. Our research group has the required expertise to carry out this project in all aspects. The HPC technologies required to undertake the project have been developed in house (with support from two ERC grants and Academy of Finland Center of Excellence, but all for astrophysics research) Our

group members are the world-leaders of modelling the solar magnetic activity We have developed the SGS and LES approaches of magnetized plasmas over decades with well-documented results. With the support from our collaborators, especially important in developing the AI techniques, we will, with no doubt, be able to successfully carry out this ambitious project and all of its subtasks. In addition to these collaborators, we have an extensive international network interested in utilizing our HPC tools, including research groups from Austria, Germany, India, Sweden, Switzerland, Taiwan, and several groups from the United Kingdom (University of Exeter, Glasgow, and Newcastle Upon Tyne), to be added with industrial collaborators.

1.2 Research questions and/or hypotheses

Our main hypothesis is that by creating an AI framework for extracting actionable insights from high-fidelity data[2] we can formulate a physically plausible SGS model. Based on it, we further hypothesize that we can develop an equally accurate as DNS-type models ELES simulation tools. As they are based on filtered equations, they will form a novel, computationally cheaper, class of simulation tools. This framework aims to capture all the relevant small-scale physics associated with a turbulent hydromagnetic environment including the physics-driven coefficients describing the turbulent transport in such systems. In particular, the inherent inhomogeneities and anisotropies occurring at these scales have to be accounted for, as well as further anomalies that have been notoriously difficult to model. Our hypotheses can be mathematically formulated as follows. Specifically, we will replace the equations nominally used to describe the bulk behaviour of plasma (know as the magnetohydrodynamic (MHD) differential equations) – solved in DNS – by a filtered formulation of the key plasma properties in this physics setup.

$$\frac{\partial \overline{\boldsymbol{B}}}{\partial t} = \nabla \times \left(\overline{\boldsymbol{U}} \times \overline{\boldsymbol{B}} + \overline{\boldsymbol{\mathcal{E}}}\right) + \eta \nabla^2 \overline{\boldsymbol{B}}, \tag{1}$$

$$\overline{\rho}\frac{D\overline{\boldsymbol{U}}}{Dt} = -\boldsymbol{\nabla}\overline{p} + \boldsymbol{\nabla}\cdot(2\nu\overline{\rho}\overline{\mathsf{S}} - \overline{\boldsymbol{\mathcal{R}}} + \overline{\boldsymbol{\mathcal{M}}}), \tag{2}$$

$$\overline{\rho}\overline{T}\frac{D\overline{s}}{Dt} = -\boldsymbol{\nabla}\cdot(\overline{\boldsymbol{F}}_{\rm rad} + \overline{\boldsymbol{\mathcal{F}}}) + \overline{Q}_{\rm diss}, \tag{3}$$

$$\frac{D\ln\overline{\rho}}{Dt} = -\boldsymbol{\nabla}\cdot(\overline{\boldsymbol{U}} + \overline{\boldsymbol{\mathcal{D}}}). \tag{4}$$

The equations describe the time evolution of the magnetic field (1), momentum (2 Navier-Stokes equation for viscous fluids), entropy (3), and continuity (conservation of mass, 4). The overbar indicates filtering. p is the pressure, ν and η are the microscopic diffusivities, $\overline{\mathsf{S}}$ is the traceless rate of strain tensor, $\overline{\boldsymbol{F}}_{\rm rad}$ is the radiative flux, and $\overline{Q}_{\rm diss}$ represents the dissipative heat sources, while $\overline{\boldsymbol{\mathcal{E}}}$, $\overline{\boldsymbol{\mathcal{R}}}$, $\overline{\boldsymbol{\mathcal{M}}}$,$\overline{\boldsymbol{\mathcal{F}}}$ and $\overline{\boldsymbol{\mathcal{D}}}$ are the electromotive force, Reynolds stresses, Maxwell stresses, turbulent heat flux, and turbulent mass flux, respectively, calculated over the filter scale. The latter five quantities, defined by the correlations of small-scale fields, are the SGS components to be modeled with ML.

Although physics-based models and physical symmetry constraints of the environment (e.g., gravity and rotation) are proposed for many of these components, their effectiveness is limited. Here, the purpose of the AI technique used is to learn new models that remain unchanged in these symmetries. In this part of the project, we aim to expand our initial DL results (see Figure 2) to more complicated scenarios e.g., including more complex physics **This moves beyond more traditional approaches**, with the main goal to develop **evolution equations** for five plasma quantities over the filter scale. The AI methods will then address their non-local and non-linear interactions with the resolved, now filtered, fields [31]. Once fully implemented, these filtered plasma parameters will be

[2]here, high-fidelity refers to data that is created by high-resolution, expensive DNS simulations.

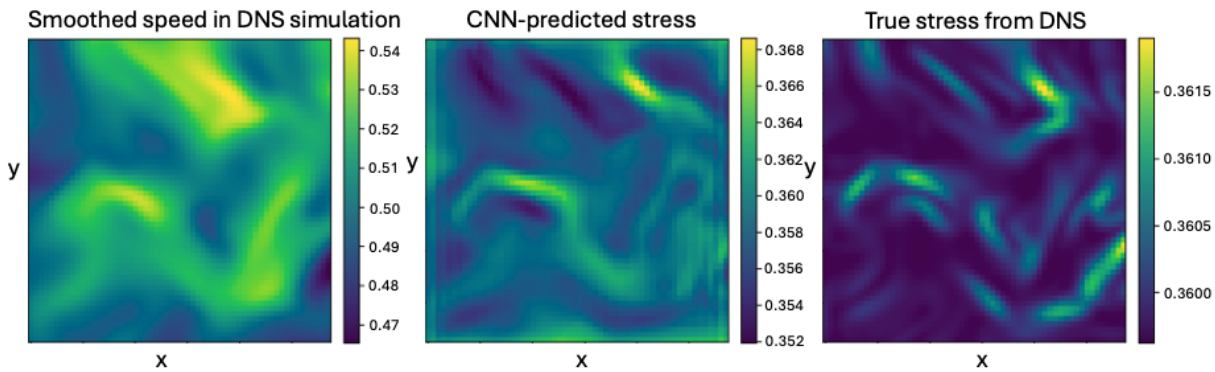

Figure 2: Proof-of-concept Taylor-Green vortex DNS simulation results (smoother mean DNS flow field shown on the panel on the left, one of the generated Reynolds stress components shown in the panel on the right) with the offline-trained AI framework and the reconstructed stresses with a U-Net type Convolutional Neural Network with symmetric upscaling and downscaling blocks and skip connections [21]. Image credit: Niki Loppi, NVIDIA.

compared and evaluated against their DNS-type counterparts. The performance of our novel models will be measured with respect to the computational time and energy consumption compared to current methods. The accuracy of our filtered parameters will also be calculated using conventional approaches (e.g., optimising the root-mean-squared error of parameters), to allow for easy comparison with current models and potential competitors.

1.3 Special objective of call

The MomEnt project addresses the following relevant aspects of the call:

- It utilises the LUMI-G supercomputer for top-notch research in fluid dynamics, relevant for many other scientific fields, such as astro-, space-, and fusion plasma physics. PI's group has actively used the LUMI supercomputer from the very beginning of its history: First we acted as pilot users of LUMI before its inauguration, and now we are the biggest individual group utilizing it from Aalto University. We have received several standard project allocations, and extreme-scale access allocations both through the Finnish and European quota.
- It builds on active collaboration with the NSF-Simons AI Institute for Cosmic Origins (Cosmic AI, https://www.cosmicai.org/), especially with the Artificial Intelligence Team Lead Prof. Arya Farahi from University of Austing, Texas, US. Collaboration with Prof. Farahi concentrates especially on the improvement of the proof-of-concept AI tools to take into account the explainability aspect of them. This is vitally important goal to be able to build physically plausible SGS models with the help of the AI tools.
- It integrates recently developed HPC tools (during ERC grants UniSDyn and SYCOS, and EuroHPC Inno4Scale project NEOSC) into a diverse HPC ecosystem, where all the building blocks together enable high-profile and societally important computational research in pre-exascale supercomputers and beyond. Furthermore, our tools automatically optimize for utilizing the heterogeneous resources as efficiently as possible, thus significantly contributing to sustainable and green computing. Furthermore, the tools and methods developed in this project are perfectly suitable to LUMI AI factory.

As is presented in Figure 3, our HPC workflow scales up in LUMI-G nearly perfectly both in weak and strong scaling scenarios.

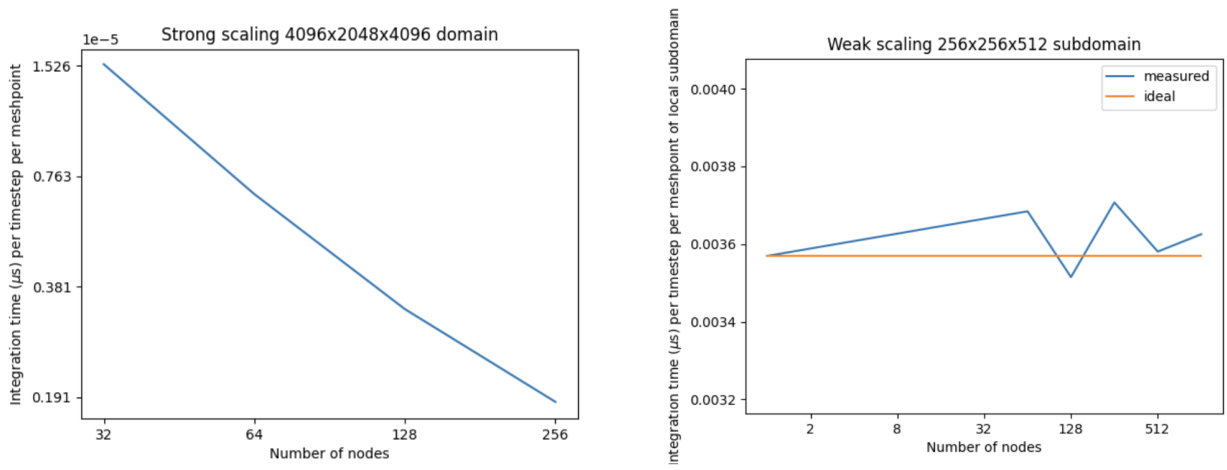

Figure 3: Scaling results of the full HPC workflow (PC-A and AI tools implemented through ODOP)in LUMI-G. The used setup us spherical turbulent convection with a sub-domain size 256x256x512 in the weak scaling case representing the planned simulation runs in the GPU-DNS concept phase [28].

2 Impact

2.1 Outcomes and impacts of research within academia

Our project aims to transform plasma simulation tools to the next level by creating novel AI methods for modeling magnetized plasmas,significantly improving computational speed and reducing computational cost. The utilization of the developed AI methods widens plasma simulations' applicability in research, spanning from solar phenomena to astrophysical and technological domains. Incorporating explainable AI (XAI), we will unlock and clarify complex plasma interactions within these systems, enhancing scientific understanding and innovation. This blend of transformative simulation and AI models, uncertainty quantification methods, and deep insights gained through explainable AI promises a paradigm shift in plasma physics research.

2.2 Effects, impact and interaction beyond academia

This project democratizes complex magnetized plasma simulations, currently limited by high computational costs and storage requirements. This project will engage a broader research community, facilitating the integration of plasma simulations with empirical research and the development of practical applications, such as space weather prediction, which require real-time data and high accuracy. Collaboration between leading international groups in AI methods and plasma simulations will foster a transdisciplinary community skilled in both physics and AI/ML. This workforce is vital for academia and industry alike. Important aspect is to democratize access to AI and plasma physics for diverse audiences, fostering inclusivity and equipping participants with cutting-edge skills essential for future scientific and engineering challenges.

3 Implementation

3.1 Work plan and schedule

Our project developes three novel concepts; the work plan can be divided into phases based on them. A detailed workplan is presented below and summarized in Figure 4.

Months 1–6: Integration of various components, their performance and scalability [GPU-DNS] – PD: development, testing and verification of the solar convection dynamo model in spherical coordinates using PC-A solver and testing and validating the DL concept in simple known scenarios PS: integration, performance, and scalability of PC-A and DL simulation-in-the-loop concept

[AI-IN-LOOP] PI: leading the integration and securing the computing resources at CSC and in LUMI-G.

Months 6–18: Extending the DL framework to magnetised plasmas – PS: Implementation of the novel LES-DNS simulation framework to PC-A (see Section 3.2 for details of the research methods) begins [LES-DNS]. PD: extension and required development and optimisations of the DL framework to the desired solar convective dynamo takes place. PD makes the first visit to University of Texas [AI-IN-LOOP]; PI: leading the novel LES solver implementation, contributing to the AI methods development, and securing further computational resources.

Months 18–24: Full next-generation plasma simulation chain [LES-DNS] is in place, tested, verified, and optimised – PS: verification and optimisation of the [AI-IN-LOOP] production pipelines. Postdoc: testing, verification, and optimisation of the convective dynamo PC-A part of the [LES-DNS] chain. PD: optimisation and further development of the AI part if necessary. PI: verification and optimization of the [LES-DNS] concept, and securing computational resources.

Months 24–36: Production runs with the full novel LES-DNS chain with unprecedented resolution of the solar convection dynamo setup. Analysis of the final results, assessment of the accuracy and validity of the developed [AI-IN-LOOP] concept. Main dissemination of the results. PD: main responsibility for running and analysing the results. PD makes the second visit to University of Texas. PS: dissemination of the methods and securing the production runs operability. PI: participating in the design of the production runs, data analysis, and dissemination of the results.

3.2 Research data and material, methods, and research environment

We will use the following **research methods**:

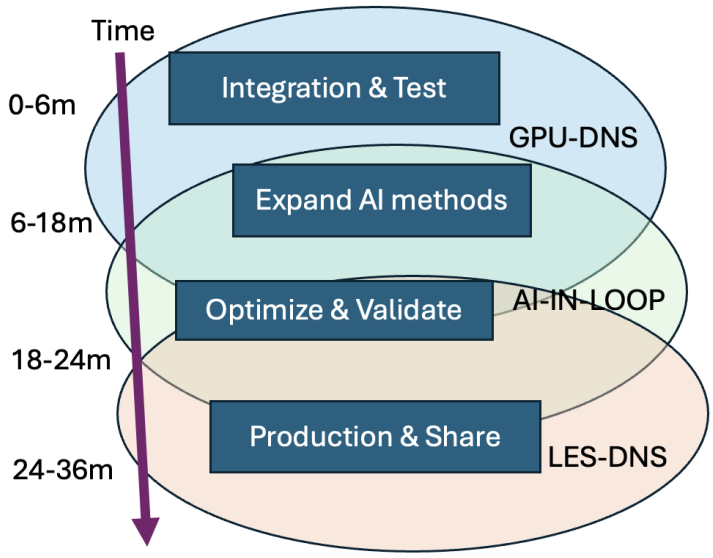

Figure 4: Timetable

- The public domain stencil library Astaroth will be the main computational engine employed during the project [26, 25].
- Astaroth is interfaced to Pencil Code, which both orchestrates the GPU computations and performs some recurrent diagnostics on the fly, utilising the CPU cores that are not involved in stencil computations through multithreading with OpenMP (PC-A; [27]).
- The ODOP framework [23] is utilised to perform further data analysis and movements, maximising the resource usage on the reserved nodes, optimising for the quality of the data products, and minimising the data storage needs on the HPC system.
- The AI development platform used is NVIDIA's PhysicsNeMo, an open-source framework for physics-based AI models. This framework uses various techniques needed to achieve our objective including, but not limited to, physics-informed neural networks and Fourier neural operators to create an architecture for analyzing MHD and differential equations as well as plasma data
- NVIDIA Warp framework will be used to build accelerated and differentiable simulation code.

- Additionally, NVIDIA's TorchFort library will be used to integrate our DL model with the DNS tool (PC-A), enabling real-time training and inference. We will apply both physical and numerical constraints to ensure the generalizability of the DL model.

The main **research environment** is the Department of Computer Science (hereafter Aalto/CS) at the Aalto University School of Science. With over 40 professors and more than 400 employees, the department is the largest Computer Science unit in Finland. It is ranked among the top 10 Computer Science departments in Europe and among the top 100 globally. ICT and digitalization are among the seven key research areas at Aalto University. In addition to standard resources provided by the department, there is support for midrange HPC and data analytics through the Science-IT project of the Aalto University School of Science. Science-IT and its flagship HPC unit Triton, are a part of the national Finnish Computing Competence Infrastructure (FCCI). Currently, Triton provides 9600 computing cores and over 30 servers for high-end GPU computing. For indepth research methods and software development, Science-IT provides support from research software engineers. Researchers also have free access to national CSC resources, including the supercomputers Puhti and Mahti, as well as the cloud resources Rahti and CPouta. The teams also have access to Europe's most powerful GPU-based supercomputer LUMI through competitive computing time allocations via the Euro-HPC consortium. The consortium has an outstanding track record of obtaining competitive computational resources through the Partnership for Advanced Computing in Europe (PRACE), EuroHPC, and CSC.

We produce mainly **simulated data** in various formats (HDF5 and binary/ASCII files), Fortran or C binaries, and AI models in suitable formats. All formats for data are highly standardised and described in the Pencil Code and Astaroth user manuals. A minimum amount of system states (in binary or HDF5 format) will be kept for allowing re-starts of the most important models. The main emphasis is on performing various types of data analyses and reductions on the fly (for example, various kinds of averages or power spectra) to provide data in a more compact and, hence, more manageable form, especially when high cadences are needed.

3.3 Risk assessment and alternative implementation strategies

This is a pioneering effort to use AI and DL for describing highly non-linear and non-local processes, and is therefore of high risk. If computationally affordable and sustainable simulation tools can indeed be formulated and developed this way as we hypothesize, this will mark a very significant milestone in computational science and engineering in general.

To achieve our objective using our AI implementation, there are many risks that need to be considered and mitigated. The main risk that can hinder this project is the potential for delays in securing computing resources. We can mitigate such delays with our already secured initial allocations of GPU time. Throughout the three-year project we will proactively apply for additional resources to ensure uninterrupted progress.

Minor risks that could potentially be an obstacle include the complexity of developing this state-of-the-art stable and generalisable SGS model for magnetised plasmas. To address this, we will adopt an iterative approach, allowing flexibility in model adaptation without compromising the project's overall objectives. Additionally, the prescribed LES approach is expected to yield significant speedups, making simulations more computationally feasible.

PI's group has extensive, world-leading expertise in high-performance computing and plasma modelling, ensuring that we can effectively handle large-scale data and computational demands. The GPU-optimised solver (Astaroth) is already in production, reducing implementation risks, and currently providing an order of magnitude performance improvement when compared to previous CPU methods. The methodologies we develop will not only support this project but also provide valuable tools for the broader plasma science community, ensuring long-term impact beyond the

immediate research goals.

3.4 Project personnel and their project-relevant key merits

The PI, **Maarit Korpi-Lagg** (over 4,800 citations and h-index of 39 according to Google Scholar), is a tenured associate professor at Aalto CS, where she leads the High-Performance Computing Lab. She was the grant holder of the ERC Consolidator grant "UniSDyn" and the ERC Proof of-concept grant “SYCOS”, which projects studied dynamo action in solar and stellar convection zones. She acted as a team leader of the Academy of Finland Centre of Excellence “ReSoLVE” (2014-2019), which studied long-term solar magnetic activity and its effects on Earth. She has acted as a PI of several competitive HPC projects (LUMI-C and LUMI-G Finnish project allocations, PRACE, DEISA/DECI, CSC Grand Challenge), and is a core developer of the Pencil Code Multiphysics Solver [8] for high-order stencil-based applications, and the initiator and project leader of the GPU-based MHD solver Astaroth.

The main research objective of our HPC Lab for the past five years has been to develop an efficient GPU-accelerated DNS tool for modelling magnetized plasmas. This project is nearing completion [26, 25, 28], with the DNS tool is in full production use in the LUMI-G partition. It has also been ported and passed technical inspection in various other supercomputers across the globe. We have been successful in obtaining large amounts of computing time at various sites. The catch here is that, while we can now go to unprecedented resolutions from the compute point-of-view, the amount of data produced by the simulations is growing too large (soon approaching Petabyte scale) to be stored in the HPC systems. In practise, the exploding amounts of data prohibit us to produce simulations at the pace we otherwise could. This kind of approach to computing will face a dead-end in the near future. Our HPC Lab has developed tools to overcome the data analysis bottleneck challenges, e.g. utilizing streaming of data from the memory of the main application of the data analysis tools on the fly, and these will be utilized here.

All these computational (and therefore sustainability) challenges we are currently facing have led us to consider the novel concept of engaging machine learning to make use of the masses of data that we currently produce, and to develop this way the next-generation modelling tools that utilize the learnt SGS models to infer the smallest-scale effects. We successfully obtained seed funding from the FARIA Global US Networking Fund of the Ministry of Education and Culture. During the project, we have actively searched for collaborations in AI, and this is now bearing fruit. For example, we have now established collaboration with the CosmicAI NSF-Simons Foundation Centre through Prof. Arya Farahi (University of Texas at Austin, coordinator of CosmicAI). We also applied for a collaborative project from the NVIDIA AI Technology centre and were awarded three person months of research engineer support to develop and apply the necessary AI methodology. The initial deep convolutional neural network has been developed, trained, and tested with data dumped to disk from a low-resolution fluid simulation with more simple physical processes. The LES-framework for the new simulation toolbox has been designed, but for its implementation, testing and verification we need further personnel resources.

The project personnel will consist of a postdoctoral researcher (PD) and a postgraduate student (PS). The **PD, to be recruited**, must have a doctoral degree in astrophysics and a strong background in computational science. The PD’s tasks include running the simulations and developing, supervised by the PI, the SGS models and the new LES solver. The PD will collaborate closely with Prof. Farahi to transform the AI tool data into knowledge that can be formulated into SGS models. **The PS in computer science has been recently recruited (MSc Touko Puro)**; the thesis topic is "Heterogeneous Exascale Scientific Computing Pipeline with Domain-Specific Language Compiler Techniques". Touko’s tasks shall include all the development and optimization tasks related to the simulation, AI, and data analysis tools, and the construction of the new LES solver. Touko is already well-integrated into our HPC lab and is intensively collaborating with Prof. Truong’s group, Dr. Loppi,

and Prof. Heljanko's team at UH.

3.5 Collaborators

Prof. Arya Farahi, University of Texas, Austin, Tx, US: Prof. Farahi is an Assistant Professor in the Statistics and Data Science department, and his research is focused on advancing research at the intersection of artificial intelligence, machine learning, and scientific applications. His expertise lies in developing explainability and uncertainty quantification methods for AI models, particularly those designed for complex scientific domains such as astronomy and solar physics. His support for the project is vital in the development of explainable AI methods and applying them to complex astrophysical flows.

Prof. Keijo Heljanko, Department of Computer Science, University of Helsinki: Prof. Heljanko is a Professor of Parallel and Distributed Data Science. He is researching Big Data processing using parallel algorithms that employ both CPUs and GPUs. He is the director of the Data Science Master's Program at the University of Helsinki and FCAI representative in the joint technology center of NVIDIA and FCAI (NVAITC) accelerating research, education and adoption of artificial intelligence in Finland. During the most recent years, he has worked intensively on transforming different types of Big Data processing algorithms to parallelize efficiently on GPUs. Prof. Heljanko will crucially support this project in the extremely challenging data analysis tasks and workflows.

Dr. Niki Loppi is an AI/HPC solutions architect at **NVIDIA Helsinki**, helping academic researchers to leverage NVIDIA's technology stack through the NVIDIA AI Technology Center program. Prior to joining NVIDIA, he worked as a researcher in the Department of Aeronautics at Imperial College London, where he also obtained his PhD in Computational Fluid Dynamics. Specifically, his research focused on the development of high-order accurate numerical methods for solving large-scale incompressible fluid flow problems using massively parallel modern GPU architectures. Currently he works on physics-informed neural networks and scientific AI in general. Dr. Loppi has collaborated with the PIs group intensively during the past year, and he has developed the proof-of-concept U-net type CNN model, which we have successfully applied in collaboration to the case of Taylor-Green vortex (see Figure 2). Dr. Loppi will collaborate with the PI in identifying of the most effective neural network architecture for MHD sub-grid scale modeling.

Prof. Hong-Linh Truong, CS, Aalto University: Prof. Truong leads the group of Systems and Service Engineering Analytics (AaltoSEA) engaged in research and development of concepts and techniques for big data systems, resource management, performance monitoring, and big data analytics for complex software systems. Prof. Truong has participated in several EU projects involving HPC, Grid computing, and scientific workflows. His role in this project is to contribute to the integration of cloud services to the HPC ecosystem, specifically to the implementation of profiling and service integration, and ecosystem-aware experiment management and explainability. The PI and prof. Truong have collaborated intensively for several years. This collaboration has resulted in the development of the ODOP tool, which will be a key component to be used in the simulation work and developed further during this project.

3.6 Responsible science

There are **no ethical concerns** as the application deals with no sensitive data. Nevertheless, we will follow stringent integrity principles in accordance with the guidelines of the Finnish Advisory Board on Research Integrity, and the ALLEA European Code of Conduct for Research Integrity.

We are committed to promote **diversity, equality and non-discrimination** in all our activities. Statutory obligations set forth in the Act on Equality between Women and Men 609/1986, 1329/2014 and the Non-discrimination Act 1325/2014 also apply to our plan. We are firmly committed to promote equal opportunities for individuals to learn, acquire knowledge, participate

and make a difference. We will make our values visible in all our practices.

The products of the research will be made available immediately to the community during the project. **Fully open access publications, data, and source code policy will be adopted.** Services like Zenodo and FAIRdata IDA will be utilised for opening up data products ranging from the data to reproduce all the figures, to intermediate and selected raw data products for reproducibility and reusability of the results. All the simulation and data analysis tools are developed as open source, available and continuously maintained under GitHub and Bitbucket repositories.

Our simulation and data analysis tools are being developed to work generally on GPUs and hence are energy-efficient in comparison to standard CPU computing. Data-efficient methods in our work help to achieve sustainable storage (also avoid data waste) and energy efficiency. Furthermore, we assess if international and national travel can be avoided by using remote visits. Computational time will be used wisely to avoid too large energy consumptions.

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